

Real Time Versatile Structured Light Imaging (rtVSLI)

Testing Results Summary

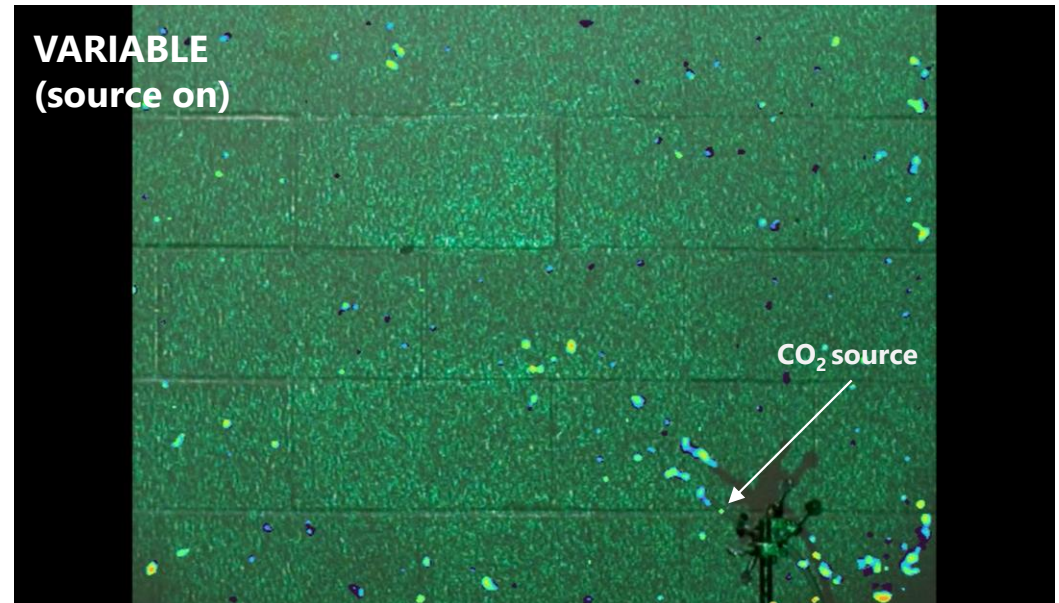
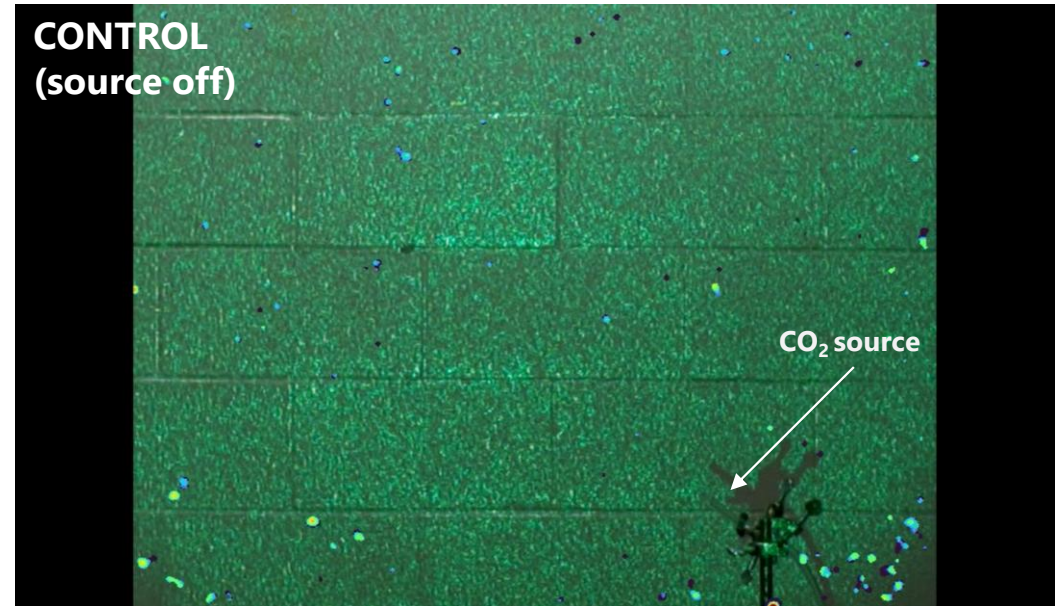
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The Battelle logo, featuring the word "BATTELLE" in a bold, italicized, sans-serif font.

CO₂ Testing

Testing Conditions

- Day 1 testing
- Distance to target: 9ft
- Laser mA: 500
- Wavelength: 520nm
- Temperature (ambient): 23°C
- Temperature (variable): 23°C
- UV Fused Silica Engineered Diffuser, 20° Circle Pattern

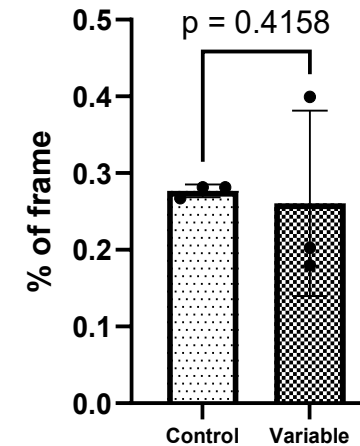


CO₂ Testing

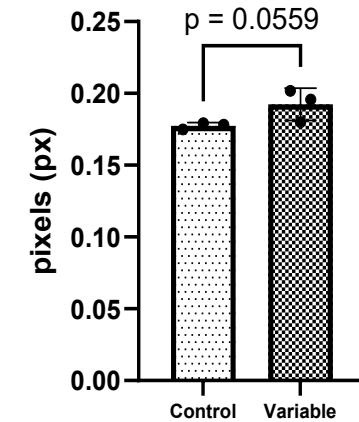
Results Summary

- Control runs: n=3
- Variable runs: n=3
- Control and variable runs were run consecutively
- p values taken from paired t-tests
- Testing had mixed results
- Largest coherent region comparison results suggest algorithm parameters used did not produce a high enough SNR to consistently perform differential analysis successfully

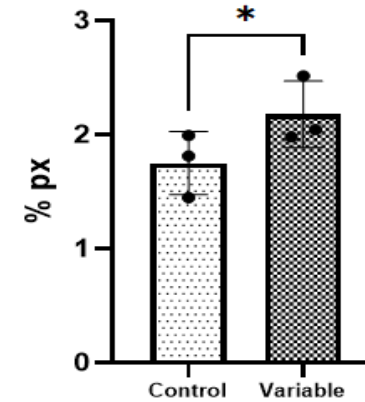
Largest Coherent Region



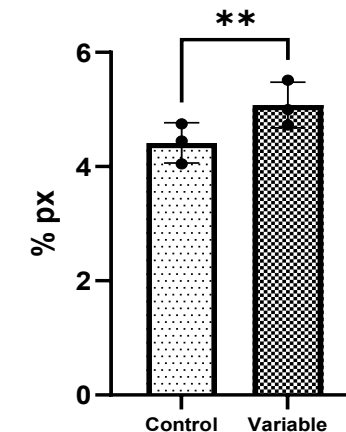
Mean Coherent Displacement



% px > Spatial Coherence Threshold



% px > I_z



CO₂ Testing

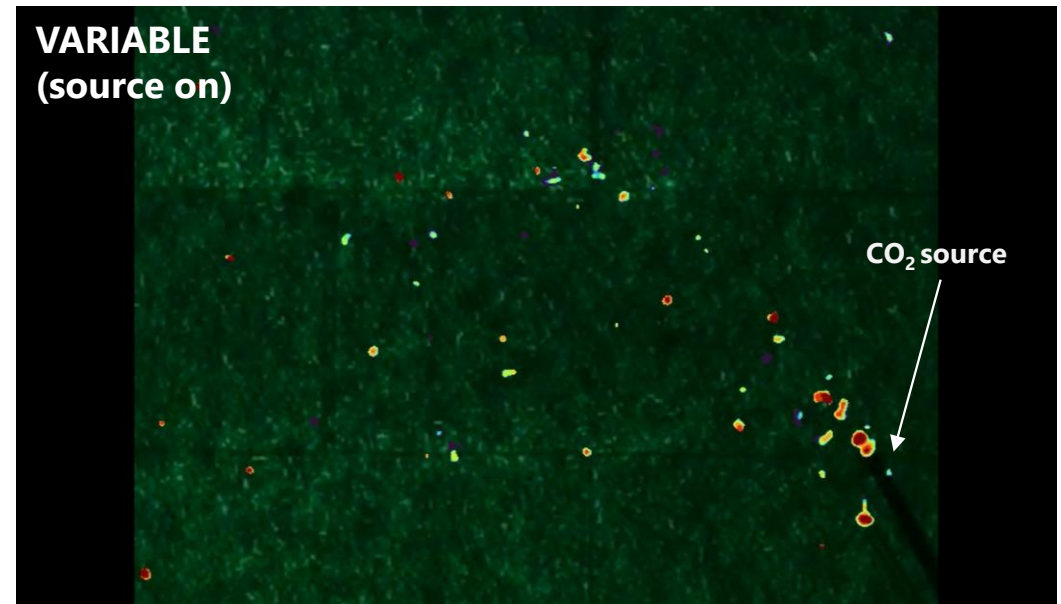
Testing Conditions

- Day 2 testing
- Distance to target: 9ft
- Laser mA: 500
- Wavelength: 520nm
- Temperature (ambient): 23°C
- Temperature (variable): 23°C
- UV Fused Silica Engineered Diffuser, 20° Circle Pattern

CONTROL
(source off)



VARIABLE
(source on)

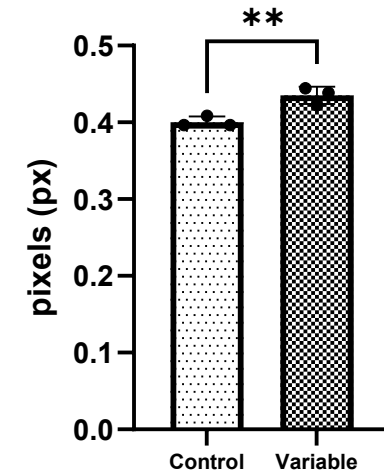
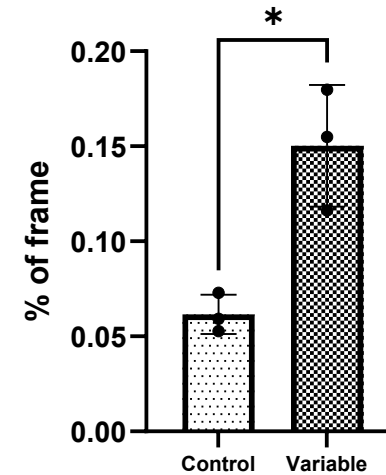


CO₂ Testing

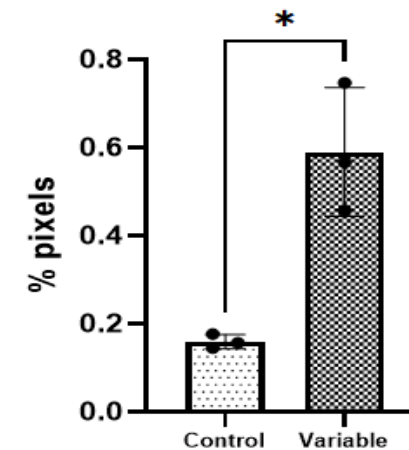
Results Summary

- Control runs: n=3
- Variable runs: n=3
- Control and variable runs were run consecutively
- p values taken from paired t-tests
- Results demonstrate successful differential measurement of denser-than-air gases in isothermal testing conditions at room temperature
- Adjustments to algorithm parameters resulted in significant improvements to SNR and differential analysis compared to 8/27/26 testing

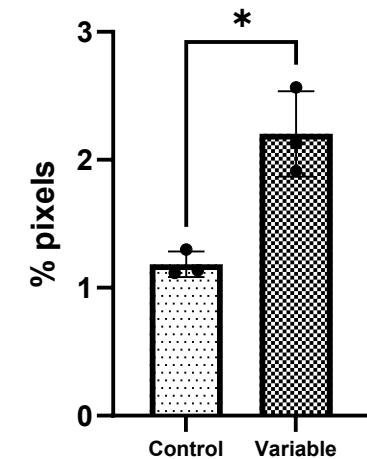
Largest Coherent Region Mean Coherent Displacement



%px > Spatial Coherence Threshold



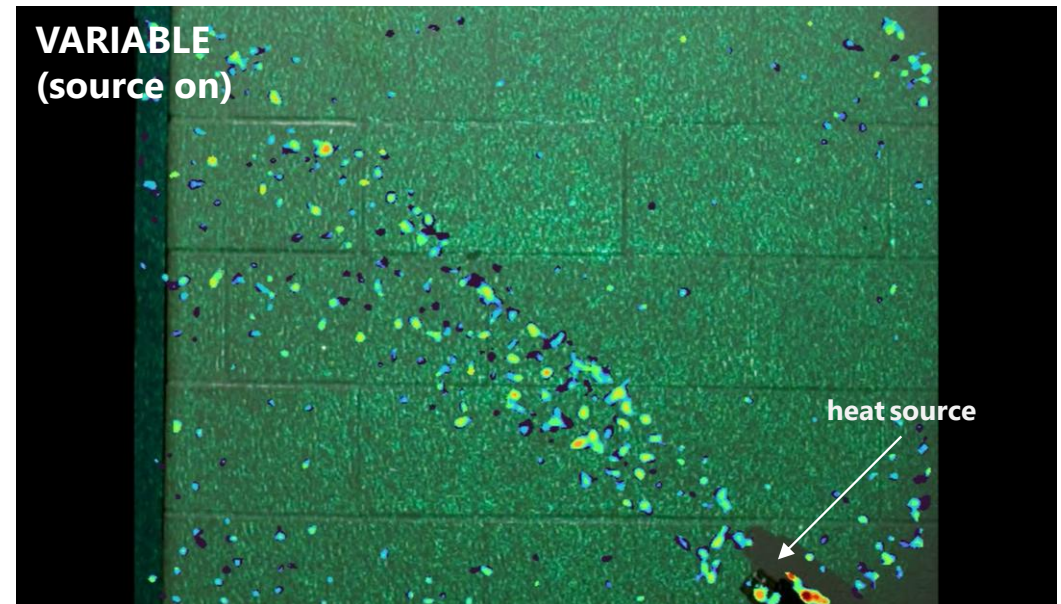
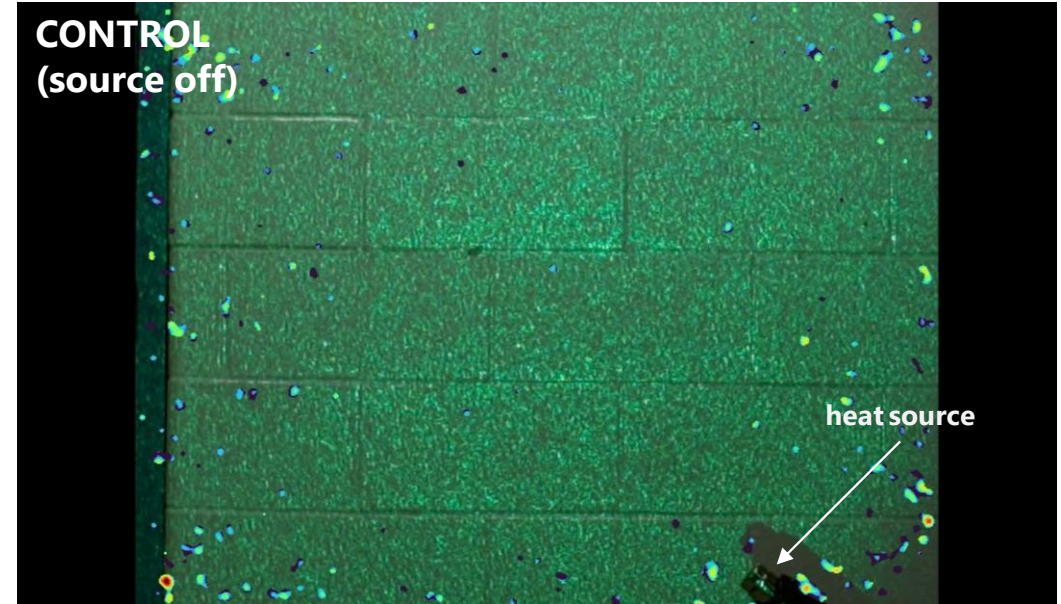
%px > I_z



Heat Testing

Testing Conditions

- Testing performed on 8/27/26
- Distance to target: 9 ft
- Laser mA: 500
- Wavelength: 520nm
- Temperature (ambient): 23°C
- Temperature (variable): 400°C
- UV Fused Silica Engineered Diffuser, 20° Circle Pattern

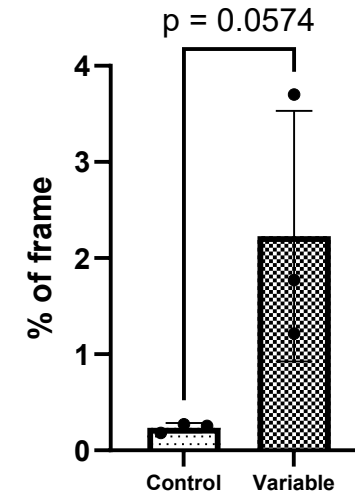


Heat Testing

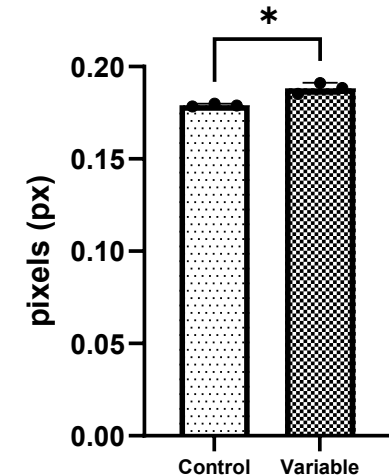
Results Summary

- Control runs: n=3
- Variable runs: n=3
- Control and variable runs were run consecutively
- p values taken from paired t-tests
- Results demonstrate successful differential measurement of thermal gradient otherwise invisible to the naked eye

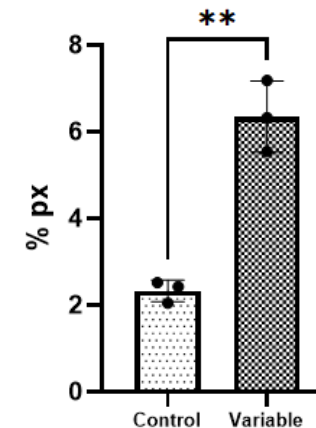
Largest Coherent Region



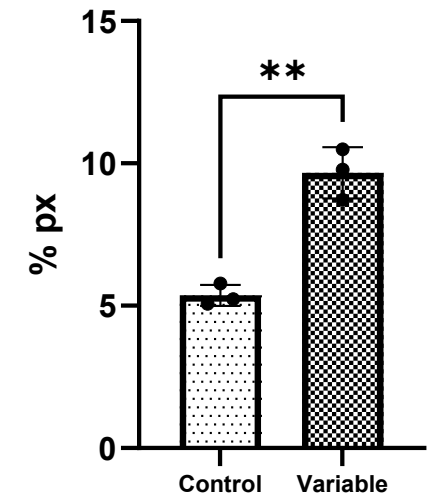
Mean Coherent Displacement



% px > Spatial Coherence Threshold



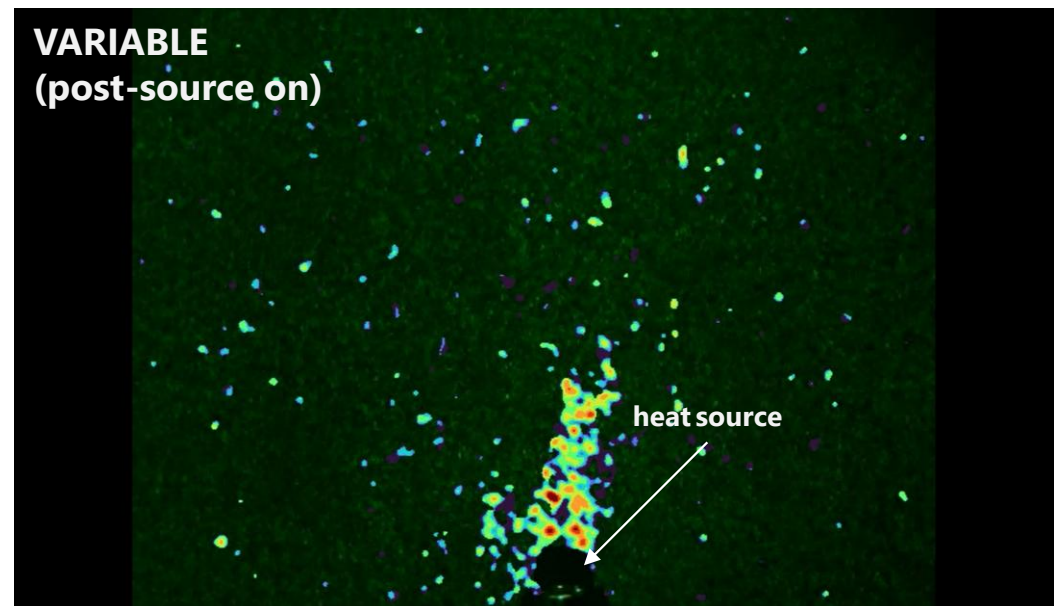
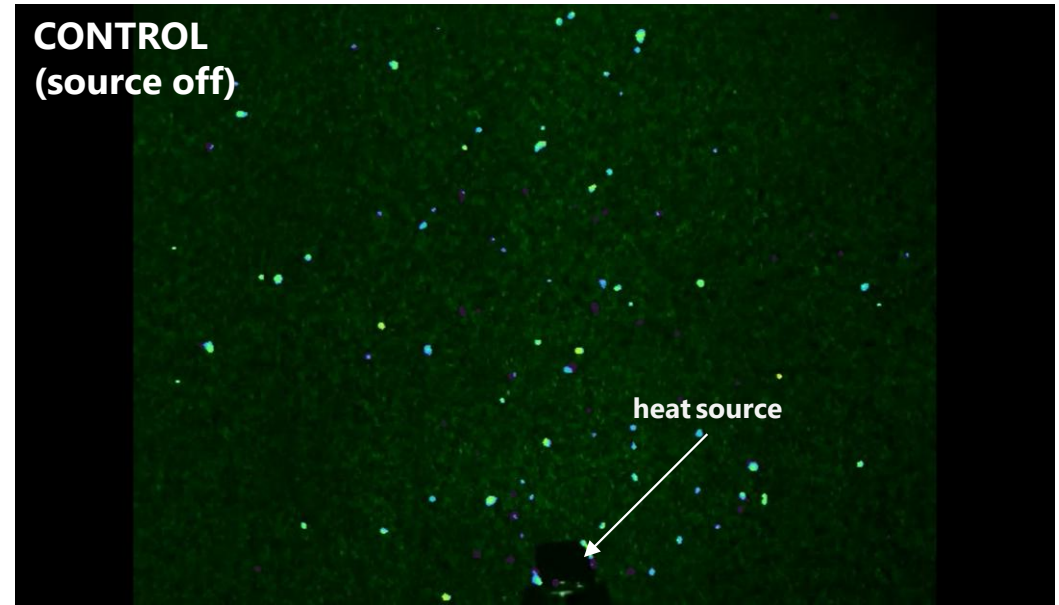
% px > I_z



Residual Heat Testing

Testing Conditions

- Testing performed on 8/27/26
- Distance to target: 5ft
- Laser mA: 353
- Wavelength: 520nm
- Temperature (ambient): 23°C
- UV Fused Silica Engineered Diffuser, 20° Circle Pattern
- Variable measurements performed after running heat source at 400°C for approx. 1 minute and then shutting off

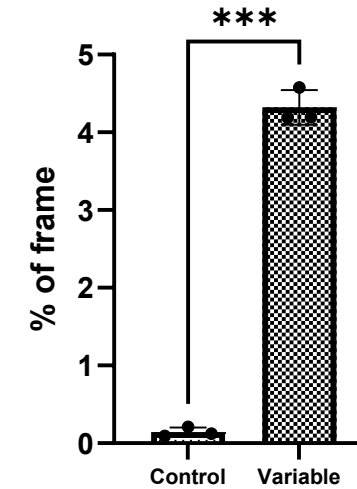


Residual Heat Testing

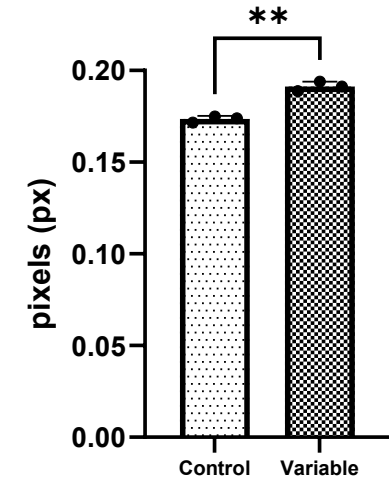
Results Summary

- Control runs: n=3
- Variable runs: n=3
- Control and variable runs not run consecutively
- p values taken from unpaired t-tests
- Captures taken without reference frames
- Results suggest successful differential measurement between frames can be achieved for sufficiently dynamic air density gradients without the need for reference frames

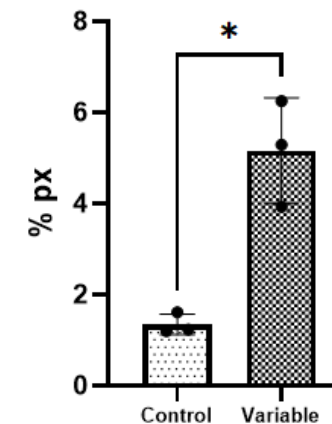
Largest Coherent Region



Mean Coherent Displacement



% px > Spatial Coherence Threshold



% px > I_z

